Natoli Tooling System TM-II Software User Requirements Specification

Table of Contents

1. Purpose and Scope 2
2. Roles & Responsibilities 2
3. System Description 2
4. User Requirements classification 3
5. User Requirements: Software 3
6. User Requirements: Hardware 6
7. User Requirements: Utility 6
8. User Requirements: Computerized System Compliance 6
9. References & Associated Documents 8
10. Document history 8
11. Appendix 8

1. PURPOSE AND SCOPE

The purpose of this User Requirements Specification (URS) is to define critical product, process and quality requirements for the Natoli Tooling Management System (TM-II).

The scope of this URS is the Natoli Tooling Management TM-II software for the use in the Process Area 5 compression area at the Newark, Delaware site. The associated Natoli Laser Vision System (LVS) has already been installed and validated and as such is not covered under this revision of the specification.

2. ROLES & RESPONSIBILITIES

Table 1 URS preparation team

Role	Name/Department	Responsibility
Project lead	Stephanie Sido	Review URS
Validation Lead	Jason Crowder	Author URS
Subject Matter Expert	Jeff Olenderski	Review and Approve URS
Owner	Reggie Smith	Review and Approve URS
QA	PJ McAllister	Review and Approve URS

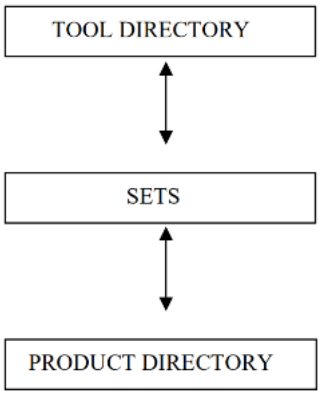
3. SYSTEM DESCRIPTION

An overview of the intended use of the Natoli TM-II software is as follows:

Tool Management II (TM-II) provides tablet manufacturers with tool control and inspection features that will improve product quality and consistency, eliminate waste, and also help ensure compliance with GMP record keeping requirements. Tool Management II can help tablet manufacturers effectively manage their tool sets in many ways including: tool storage; inspection and tolerance reporting; usage monitoring; identifying tool location; maintaining tool drawing files; printing reports; efficiently creating purchase orders for additional or replacement tool sets.

The Tooling Directory is the main table upon which the system is based. Tools can be combined into Sets. Sets of tools can be inventoried for Products. Tools do not have to be part of a Set. A tool can, however, be in more than one Set.

The Set Table provides the link between the Tool Directory and the Product they are inventoried to produce.



4. USER REQUIREMENTS CLASSIFICATION

The requirements have been classified with respect to critical aspect as CQA/CPP and/or cGMP or non GMP.

Direct Impact: CQA/CPP and/or cGMP means that the requirement needs to be met and included in the qualification in order to fulfill critical aspect and intended use.

Non direct impact: No GMP impact or No product impact. The requirements need to be met but not to be included in the qualification. Requirements can be handled outside GVLMS through commissioning.

5. USER REQUIREMENTS: SOFTWARE

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
5.1.	Natoli TMII Software	
5.1.1.	The system shall be able to manage a tool directory that contains entries for punches and dies used in production.	Direct

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
5.1.1.1.	The tool directory shall contain: • Identifyer information for each tool i.e. unique tool number, part type, tool type, etc. • Supplier information for each tool i.e. supplier ID number, P.O. number, etc. • Tooling Material information for each tool i.e. steel type, max tip force, etc. • Tool Drawings for each tool • Tool Tolerance information for each tool i.e. length, cup depth, tip diameter, etc. • Ovewrall Tool status information for each tool i.e. last inspection, last maintenance, quarantine status, etc.	Direct
5.1.1.2.	The system shall have the ability to add, edit, or remove new tools to the tool directory.	Direct
5.1.1.3.	The system shall permit tool data to be imported directly into the Tooling Directory	Non-Direct
5.1.1.4.	The system shall have the capability to search the tool directory list by text query.	Non Direct
5.1.2.	The system shall be able to manage a set directory which contains an entry of all tools contained in each set.	Direct
5.1.2.1.	The system shall have the ability to add, edit, or remove Sets.	Direct
5.1.3.	The system shall be able to link Tools from the Tool Directory to Tool Sets in the Tool Sets Directory.	Direct
5.1.4.	The system shall be able to manage a Product / Tablet Directory that contains an entry for every product or tablet	Direct

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
5.1.4.1.	The Product directory shall contain: o Product overview information o Recommended default limits for tools associated with the product o Product tolerances as related to tool wear	Direct
5.1.5.	The system shall be able to link Tool Sets from the Set Directory to Products in the Product / Tablet Directory.	Non-Direct
5.1.6.	The system must be able to interface with the Natoli Laser Vision System.	Direct
5.1.7.	The system shall have the ability to maintain a history of tool usage which includes at a minimum the total number of tablets produced per tool.	Non-Direct
5.1.8.	The system shall be able to generate a maintenance report for each Tool.	Direct
5.1.9.	The system shall be able to generate a usage report for each tool	Direct
5.1.10.	The system shall be able to generate a Tool Summary report which includes at a minimum a high level overview of the activity for each Tool or Tool Set	Direct
5.1.11.	The system shall have the ability to quarantine Tools and Tool Sets as needed.	Non-Direct
5.1.12.	The system shall have the ability to perform tool inspections in conjunction with a validated tooling measurement system.	Direct
5.1.13.	The system shall be able to display results from completed tool inspections.	Direct

6. USER REQUIREMENTS: HARDWARE

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
6.1.	The system shall utilize a windows based computer running Windows 10 or newer	Non-Direct
6.2.	The system shall have a minimum of 64MB memory.	Non-Direct
6.3.	The system shall have a minimum of 800MB available disk space	Non-Direct
6.4.	The system shall have a DVD drive and a USB port.	Non-Direct

7. USER REQUIREMENTS: UTILITY

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
7.1.	The system shall hutilize a safe power source of 120VAC power.	Non-Direct

8. USER REQUIREMENTS: COMPUTERIZED SYSTEM COMPLIANCE

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
8.1.	The system shall untilize the Astrazeneca Active Directory for user authentication.	Direct
8.2.	The system shall use AstraZeneca USUNPCN domain standard to define password length, history, and lockout	Direct

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
8.3.	The software shall provide user access control capability supporting user logon, password, and functional roles or user access levels. At a minimum, the machine should be capable to configure four user groups (Operator, Mechanic, Tech., and Engineer). AstraZeneca to provide guidance on user rights as part of the design review process.	Direct
8.4.	The system shall automatically logout users after a defined period of time	Direct
8.5.	The system shall be capable of producing accurate and complete copies of electronic records on paper. [11.10(b)]	Direct
8.6.	The system shall be capable of producing accurate and complete copies of electronic records in electronic form. [11.10(b)]	Direct
8.7.	The system shall have a documented record retention policy. [11.10(c)]	Direct
8.8.	Electronic records shall be readily retrievable throughout their documented retention period. [11.10(c)]	Direct
8.9.	The system shall contain a secure, computer-generated, time-stamped audit trail that records the date and time of operator entries and actions that create, modify, or delete electronic records. [11.10(e)]	Direct
8.10.	The system shall not obscure previously recorded information upon changing an electronic record. [11.10(e)]	Direct
8.11.	The audit trail for an electronic record shall be retrievable throughout the record's retention period. [11.10(e)]	Direct
8.12.	The audit trail shall be available for review and copying. [11.10(e)]	Direct
8.13.	The system shall ensure that only authorized users can access and perform required operations. [11.10(g)]	Direct
8.14.	Signed electronic records shall contain the following signature attributes: Printed name of signer: Date and time when signature was executed, The meaning of the signature. [11.50(a)]	Direct

URS Reference #	Requirements	Source Critical Aspect (CQA/CPP/ Direct/ Non Direct)
8.15.	The system shall display signature attributes on the displayed electronic record. [11.50(b)]	Direct
8.16.	The system shall display signature attributes on printed copies of the electronic record. [11.50(b)]	Direct
8.17.	There shall be a procedure for electronically disabling an account. [11.300(c)]	Direct
8.18.	The system shall be able to detect attempts at unauthorized use. [11.300(d)]	Direct
8.19.	The system shall be validated to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records. [11.10(a)]	Direct

9. REFERENCES & ASSOCIATED DOCUMENTS

Number	Ver.	Title
STND-0000451	8.0	Qualification and Validation Standard
SOP-0034066	8.0	General Qualification and Validation
SOP-0034104	8.0	Equipment, Utility, Facility Qualification
STND-0000429	5.0	Standard for Computerised Systems

The version in the table indicate the current version of the document at the time of issue of this URS.

10. DOCUMENT HISTORY

Version	Description of change
1.0	• New User Requirements Specification

11. APPENDIX

Not applicable for this document.

Document Approvals

Administrative Approval	Jeff Olenderski Jeffery.Olenderski@astrazeneca.com 18-Dec-2025 20:09:51 GMT+0000

Business Approval	Reggie Smith Reggie.Smith@astrazeneca.com 18-Dec-2025 20:29:17 GMT+0000

Quality Approval	PJ McAllister Jr. Pj.Mcallisterjr@astrazeneca.com 18-Dec-2025 22:22:42 GMT+0000